

Frederick A

AI & Full stack Engineer

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👤 PROFILE

Final-year B.Tech (CS/AI-ML) student with hands-on experience building full-stack web applications, Python-based ML pipelines, and real-world deployment projects. Proficient in the MERN stack, SQL/NoSQL databases, Git, and Agile development workflows. Experienced in writing clean, testable code across the full SDLC — from requirements through deployment. Strong communicator with demonstrated team leadership (150+ member organisations, national-level events) and a track record of shipping production-ready systems on deadline.

🧠 SKILLS

Languages — Python, JavaScript, C++

Frontend — React.js, Next.js, Vue.js, Angular, Tailwind CSS, Bootstrap, Material-UI, Shadcn UI

Backend & Databases — Node.js, Express, MongoDB, MySQL (SQL), SSMS

AI / ML — scikit-learn, OpenCV, MediaPipe, NumPy, Pandas, Matplotlib, LLM inference, Agentic workflows

Tools & Practices — Git, GitHub, Agile/Scrum, SDLC, REST APIs, Google Sheets API, Discord API, VS Code,

AI Tools — OpenClaw, n8n, Hermes 3, Google Antigravity, Google Stitch

Creative Design — Adobe Photoshop, Adobe Illustrator, Canva, UI/UX prototyping, Photography, Visual editing, Event branding

🎓 EDUCATION

Higher secondary 2023
Seventh-Day Adventist Matriculation Higher Secondary School
Marks: 81%

Minor in AI and Data Science 05/2025 – Present
IIT Indore

B.Tech 2023 – 2027
Kalasalingam Academy of Research and Education
Current CGPA: 7.64

🧰 EXPERIENCE

Machine learning with Python 05/2025 – 06/2025
NSIC Chennai, India

- Delivered 5+ supervised ML models across the full SDLC — from data ingestion and feature engineering to evaluation and documentation — using Python, NumPy, Pandas, and Matplotlib.
- Reduced model validation error by 89% by applying PCA for dimensionality reduction and tuning Bagging & Boosting ensembles across 12+ experiments with structured test/train splits.
- Collaborated within an Agile workflow to ship a working NLP text-classifier in under 3 weeks, applying OOPs design principles to build reusable, modular preprocessing pipelines.
- Strengthened applied knowledge of data structures and algorithms through hands-on implementation of SVM, Decision Trees, Logistic Regression, and Naive Bayes classifiers on real-world datasets.

PROJECTS

Autonomous IT Admin Agent

Hermes · Ollama · Discord API · Python

- Architected and deployed a fully autonomous AI agent functioning as an IT administrator inside a Discord server, handling real-time task execution.
- Leveraged Hermes 3 self-improvement capabilities and advanced memory management for persistent, context-aware autonomous workflows.
- Optimized inference strategy: local Ollama instances for routine operations, cloud APIs for complex architectural tasks — reducing latency by ~40%.

VigilantDrive AI — Driver Monitoring System

Python · OpenCV · MediaPipe · scikit-learn

- Developed and deployed a real-time driver drowsiness and distraction detection system using computer vision and ML techniques.
- Applied MediaPipe FaceMesh for 468-point facial landmark extraction to compute Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) in real time.
- Achieved reliable detection at 30+ FPS on standard hardware through optimized pipeline design.

College Event Website(SPARKZ) — Architecture & Management

HTML · CSS · JavaScript · CMS

- Managed and updated the full website architecture for large-scale national college fests across multiple event iterations.
- Ensured consistent branding, responsive design, and optimal UX for thousands of student visitors per event cycle.

RFID Attendance Monitoring System

Node.js · MySQL · Google Sheets API · JavaScript

- Engineered a full-stack RFID-based attendance tracking system with automated real-time updates via Google Sheets integration and a MySQL backend
- Built a multi-subject attendance engine using timetable logic to automatically map RFID scans to the correct subject based on schedule
- Developed separate student and admin dashboards with role-based access control and secure user authentication
- Designed the Node.js application layer to handle hardware RFID input, database writes, and Google Sheets sync in a single automated pipeline

LEADERSHIP & ACTIVITIES

- President, Fine Arts Club — led a 150+ member organisation with a 27-year legacy, managing budgets, vendor coordination, and a 20-person core team.
- Lead Organiser, SPARKZ National Cultural Fest — drove end-to-end execution including logistics, branding, and the official website for 7000+ attendees across multiple annual iterations.